



Southeast University

Department of Computer Science and Engineering (CSE)

School of Sciences and Engineering

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LAB REPORT NO: 02

Course Title: Operating Systems Lab

Course Code: CSE362.2

Batch: 65

Lab Experiment Name: Implementing some advanced Linux command.

Student Details

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Submission Date : 18-07-26

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Lab Report Status

Marks:

Comments:.....

Signature:.....

Date:.....

Command 1: wc

Description: Counts the number of lines, words, bytes, or characters in a file

wc -c > Counts and displays the total number of bytes

wc -l > Counts and displays the total number of lines

wc -w > Counts and displays the total number of words.

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ wc -c aaa.txt
134 aaa.txt
mahmud@mahmud-VMware-Virtual-Platform:~$ wc -l aaa.txt
4 aaa.txt
mahmud@mahmud-VMware-Virtual-Platform:~$ wc -w aaa.txt
27 aaa.txt
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 2: grep

Description: Searches for patterns within files and then prints each line that matches the pattern for which you are searching

grep -i > Searches for the pattern (case-insensitive) from standard input

grep -n > Searches for the pattern inside .txt file, ignoring case

grep -w > Searches for the exact whole word pattern in .txt file.

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ grep -i a aaa.txt
Please make a new text file.
Can you create a text file for me?
Open a blank text file, please.
Save this information in a text file.
mahmud@mahmud-VMware-Virtual-Platform:~$ grep -n a aaa.txt
1:Please make a new text file.
2:Can you create a text file for me?
3:Open a blank text file, please.
4:Save this information in a text file.
mahmud@mahmud-VMware-Virtual-Platform:~$ grep -w a aaa.txt
Please make a new text file.
Can you create a text file for me?
Open a blank text file, please.
Save this information in a text file.
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 3: sort

Description: Sorts the lines of text files in a specified order

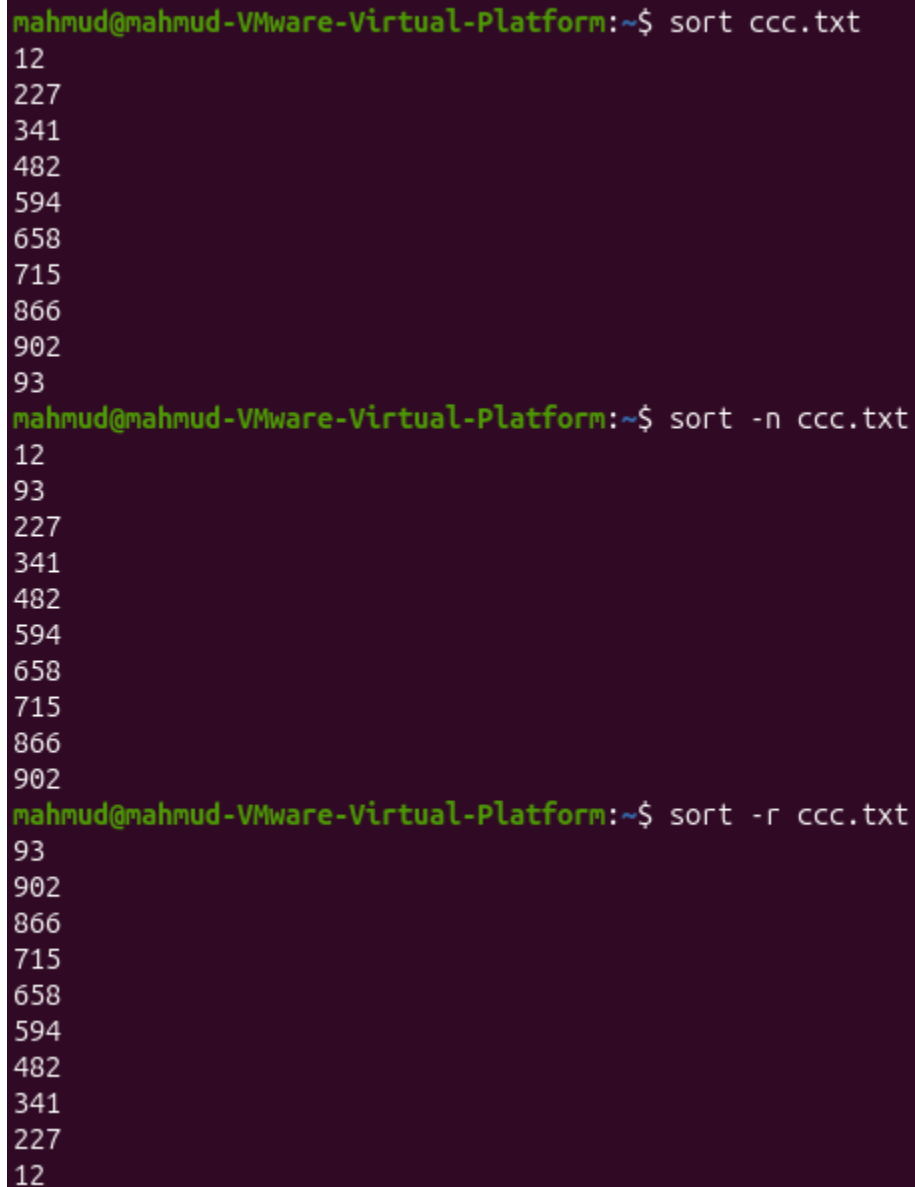
sort > Sorts the lines of a file alphabetically

sort -n > Sorts the lines of a file numerically

sort -r -n > Sorts the lines of a file numerically in reverse (descending) order

sort -r -n -o > Sorts the lines numerically in reverse and outputs/writes the result to a specified file.

Screenshot:



```
mahmud@mahmud-VMware-Virtual-Platform:~$ sort ccc.txt
12
227
341
482
594
658
715
866
902
93
mahmud@mahmud-VMware-Virtual-Platform:~$ sort -n ccc.txt
12
93
227
341
482
594
658
715
866
902
mahmud@mahmud-VMware-Virtual-Platform:~$ sort -r ccc.txt
93
902
866
715
658
594
482
341
227
12
```

```
mahmud@mahmud-VMware-Virtual-Platform:~$ sort -r ccc.txt
93
902
866
715
658
594
482
341
227
12
mahmud@mahmud-VMware-Virtual-Platform:~$ sort -r -n ccc.txt
902
866
715
658
594
482
341
227
93
12
mahmud@mahmud-VMware-Virtual-Platform:~$ sort -r -n -o bbb.txt
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 4: cmp

Description: Compares two files byte-by-byte and reports the location of the first difference

cmp > Checks two files and tells you the exact line number and byte number where they first differ.

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ cmp bbb.txt ccc.txt
bbb.txt ccc.txt differ: byte 1, line 1
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 5: comm

Description: Compares two sorted files line-by-line and outputs the data in three columns

comm > Displays unique lines of file1, unique lines of file2, and lines common to both.

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ comm bbb.txt ccc.txt
comm: file 1 is not in sorted order
comm: file 2 is not in sorted order
    482
902
866
715
658
594
482
341
227
          93
12
      715
      341
      866
      12
      594
      902
      227
      658
comm: input is not in sorted order
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 6: diff

Description: Compares two files line-by-line and lists the exact differences needed to make them match

diff > Analyzes two files and outputs the specific additions, deletions, or modifications required to make them identical.

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ diff bbb.txt ccc.txt
1,5d0
< 902
< 866
< 715
< 658
< 594
7,8d1
< 341
< 227
9a3,5
> 715
> 341
> 866
10a7,10
> 594
> 902
> 227
> 658
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 7: time

Description: Measures how long a specific command or a group of commands takes to execute
time

touch > Creates or updates a file while measuring how long the operation took

time { ... } > Runs a block of multiple commands in sequence and measures the total execution time for all of them

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ time touch bbb.txt

real    0m0.007s
user    0m0.001s
sys     0m0.003s
mahmud@mahmud-VMware-Virtual-Platform:~$
```

```
mahmud@mahmud-VMware-Virtual-Platform:~$ time { comm bbb.txt ccc.txt; diff bbb.txt ccc.txt; }
comm: file 1 is not in sorted order
comm: file 2 is not in sorted order
      482
902
866
715
658
594
482
341
227
      93
12
      715
      341
      866
      12
      594
      902
      227
      658
comm: input is not in sorted order
1,5d0
< 902
< 866
< 715
< 658
< 594
7,8d1
< 341
< 227
9a3,5
> 715
> 341
> 866
10a7,10
> 594
> 902
> 227
> 658

real    0m0.008s
user    0m0.000s
sys     0m0.006s
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 8: date

Description: Displays or calculates dates and system time

date > Displays the current system date and time

date -d > Calculates and displays a specific relative date string (like "last friday") without changing the system time

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ date
Fri Jul 17 10:36:29 PM +06 2026
mahmud@mahmud-VMware-Virtual-Platform:~$ date -d "last friday"
Fri Jul 10 12:00:00 AM +06 2026
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 9: cal

Description: Displays a formatted calendar in the terminal

cal > Displays the calendar for the current month

cal -3 > Displays the previous, current, and next month side-by-side

cal -y > Displays the calendar for the entire current year

cal [year] > Displays the calendar for a specific year (like 1971)

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ cal
      July 2026
Su Mo Tu We Th Fr Sa
                1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

mahmud@mahmud-VMware-Virtual-Platform:~$ cal -3
      June 2026           July 2026           August 2026
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6      1  2  3  4              1  2  3  4  5  6  7  8
Trash 9 10 11 12 13    5  6  7  8  9 10 11    9 10 11 12 13 14 15
14 15 16 17 18 19 20 12 13 14 15 16 17 18    16 17 18 19 20 21 22
21 22 23 24 25 26 27 19 20 21 22 23 24 25    23 24 25 26 27 28 29
28 29 30              26 27 28 29 30 31      30 31

mahmud@mahmud-VMware-Virtual-Platform:~$ cal -y
      2026
      January           February           March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                1  2  3      1  2  3  4  5  6  7      1  2  3  4  5  6  7
 4  5  6  7  8  9 10      8  9 10 11 12 13 14      8  9 10 11 12 13 14
11 12 13 14 15 16 17    15 16 17 18 19 20 21    15 16 17 18 19 20 21
18 19 20 21 22 23 24    22 23 24 25 26 27 28    22 23 24 25 26 27 28
25 26 27 28 29 30 31      29 30 31
```



```
mahmud@mahmud-VMware-Virtual-Platform:~$ cal -y
2026
    January                February                March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                   1 2 3    1 2 3 4 5 6 7    1 2 3 4 5 6 7
 4 5 6 7 8 9 10    8 9 10 11 12 13 14    8 9 10 11 12 13 14
11 12 13 14 15 16 17 15 16 17 18 19 20 21 15 16 17 18 19 20 21
18 19 20 21 22 23 24 22 23 24 25 26 27 28 22 23 24 25 26 27 28
25 26 27 28 29 30 31                29 30 31

    April                  May                  June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                   1 2 3 4                   1 2                   1 2 3 4 5 6
 5 6 7 8 9 10 11    3 4 5 6 7 8 9    7 8 9 10 11 12 13
12 13 14 15 16 17 18 10 11 12 13 14 15 16 14 15 16 17 18 19 20
19 20 21 22 23 24 25 17 18 19 20 21 22 23 21 22 23 24 25 26 27
26 27 28 29 30                24 25 26 27 28 29 30 28 29 30
                               31

    July                  August                September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                   1 2 3 4                   1                   1 2 3 4 5
 5 6 7 8 9 10 11    2 3 4 5 6 7 8    6 7 8 9 10 11 12
12 13 14 15 16 17 18 9 10 11 12 13 14 15 13 14 15 16 17 18 19
19 20 21 22 23 24 25 16 17 18 19 20 21 22 20 21 22 23 24 25 26
26 27 28 29 30 31    23 24 25 26 27 28 29 27 28 29 30
                               30 31

    October                November                December
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                   1 2 3    1 2 3 4 5 6 7    1 2 3 4 5
 4 5 6 7 8 9 10    8 9 10 11 12 13 14    6 7 8 9 10 11 12
11 12 13 14 15 16 17 15 16 17 18 19 20 21 13 14 15 16 17 18 19
18 19 20 21 22 23 24 22 23 24 25 26 27 28 20 21 22 23 24 25 26
25 26 27 28 29 30 31 29 30                27 28 29 30 31
```

```
mahmud@mahmud-VMware-Virtual-Platform:~$ cal 1971
1971
    January                February                March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                   1 2    1 2 3 4 5 6    1 2 3 4 5 6
 3 4 5 6 7 8 9    7 8 9 10 11 12 13    7 8 9 10 11 12 13
10 11 12 13 14 15 16 14 15 16 17 18 19 20 14 15 16 17 18 19 20
17 18 19 20 21 22 23 21 22 23 24 25 26 27 21 22 23 24 25 26 27
24 25 26 27 28 29 30 28                28 29 30 31
31

    April                  May                  June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                   1 2 3                   1                   1 2 3 4 5
 4 5 6 7 8 9 10    2 3 4 5 6 7 8    6 7 8 9 10 11 12
11 12 13 14 15 16 17 9 10 11 12 13 14 15 13 14 15 16 17 18 19
18 19 20 21 22 23 24 16 17 18 19 20 21 22 20 21 22 23 24 25 26
25 26 27 28 29 30    23 24 25 26 27 28 29 27 28 29 30
                               30 31

    July                  August                September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                   1 2 3    1 2 3 4 5 6 7    1 2 3 4
 4 5 6 7 8 9 10    8 9 10 11 12 13 14    5 6 7 8 9 10 11
11 12 13 14 15 16 17 15 16 17 18 19 20 21 12 13 14 15 16 17 18
18 19 20 21 22 23 24 22 23 24 25 26 27 28 19 20 21 22 23 24 25
25 26 27 28 29 30 31 29 30 31                26 27 28 29 30

    October                November                December
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
                   1 2    1 2 3 4 5 6    1 2 3 4
 3 4 5 6 7 8 9    7 8 9 10 11 12 13    5 6 7 8 9 10 11
10 11 12 13 14 15 16 14 15 16 17 18 19 20 12 13 14 15 16 17 18
17 18 19 20 21 22 23 21 22 23 24 25 26 27 19 20 21 22 23 24 25
24 25 26 27 28 29 30 28 29 30                26 27 28 29 30 31
31
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Conclusion:

In this experiment, fundamental Linux commands were practiced to understand text processing, file sorting, data comparison, execution timing, and system utilities. Gaining hands-on experience with these utilities highlights the efficiency of the Command Line Interface (CLI) over traditional graphical interfaces for managing data and tracking performance. Ultimately, mastering these core commands provides the essential foundation required to navigate and utilize the Linux operating system effectively in more advanced system-level tasks.